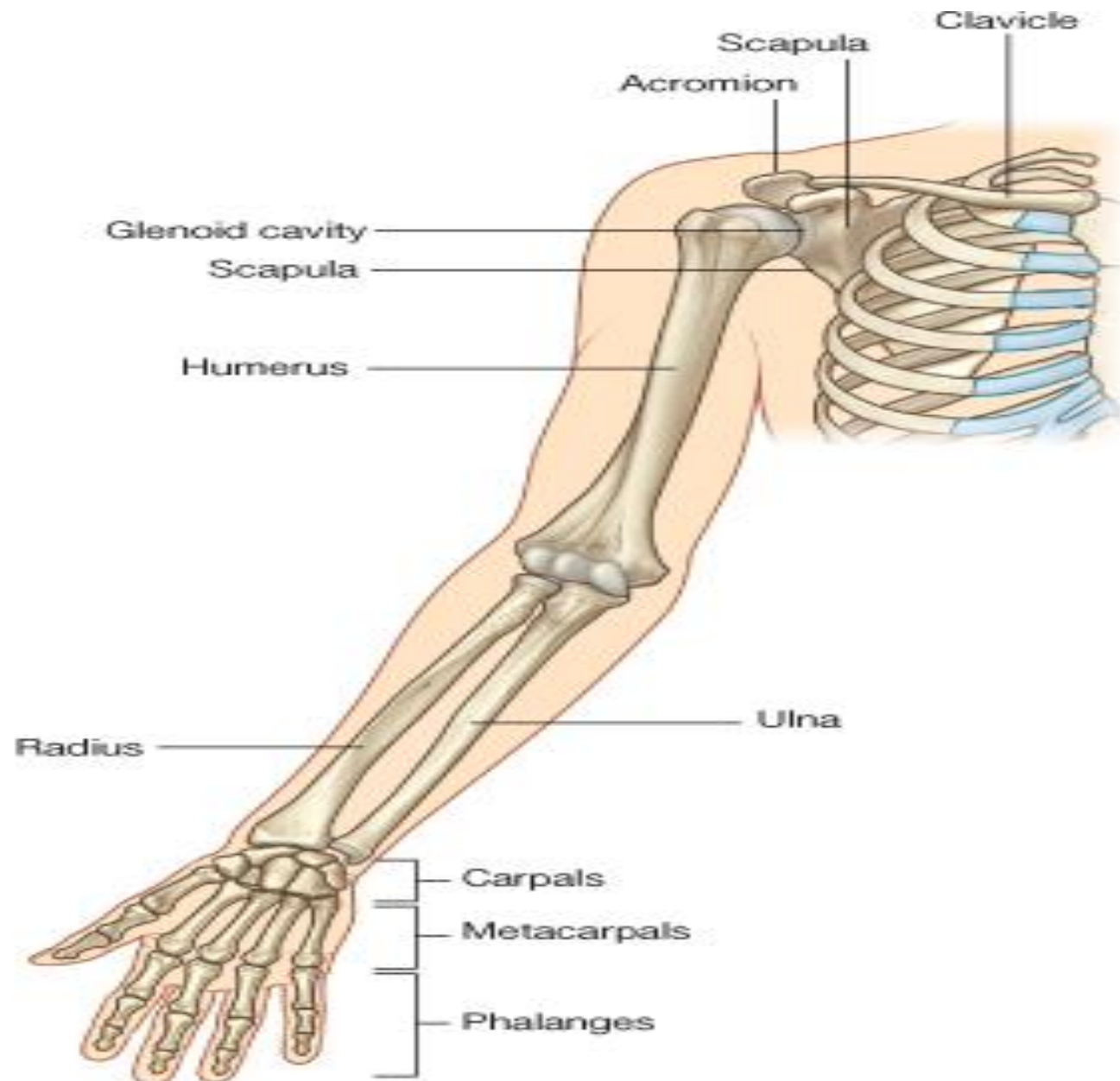


LECTURE TWO

JOINTS OF THE UPPER LIMB

JOINTS OF THE UPPER LIMB



- **Ligaments**

- **Anterior and posterior SC ligaments**

- reinforce joint capsule anteriorly and posteriorly

- **Interclavicular ligament**

- strengthens the capsule superiorly

- **Costoclavicular ligament**

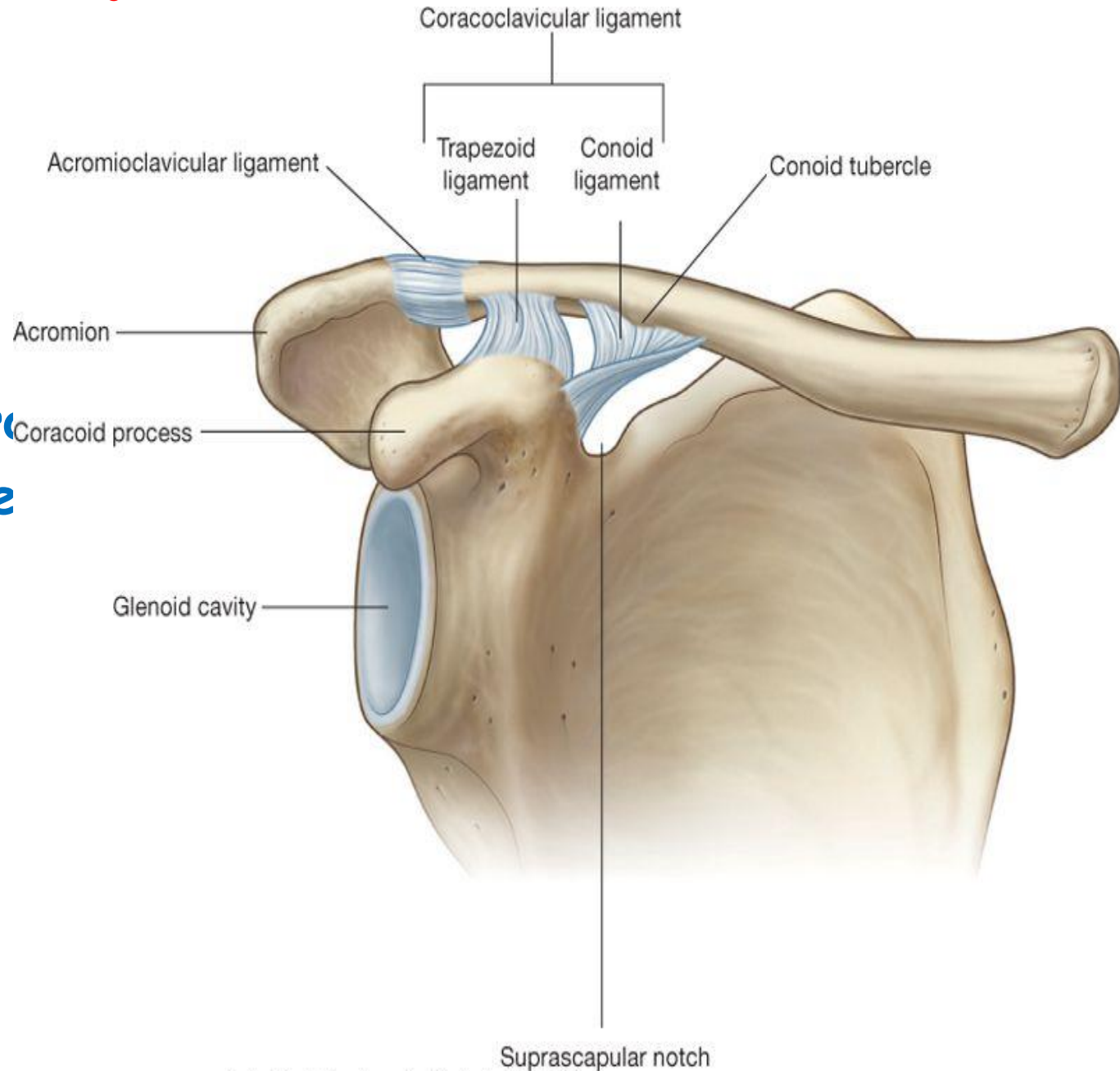
- anchors the inferior surface of the sternal end of the clavicle to the 1st rib and its costal cartilage
 - limiting elevation of the pectoral girdle

- **Blood supply:** internal thoracic and suprascapular arteries

- **Innervation:** Branches of the medial suprascapular nerve and the nerve to the subclavius

Acromioclavicular joint

- Type
- Articulations
- Cartilages (fibro)
- Articular capsule
- synovial membrane



- **Ligaments**

- **Acromioclavicular ligament**

- fibrous band extending from the acromion to the clavicle, strengthens the AC joint superiorly

- **Coracoclavicular (trapezoid & conoid) ligament**

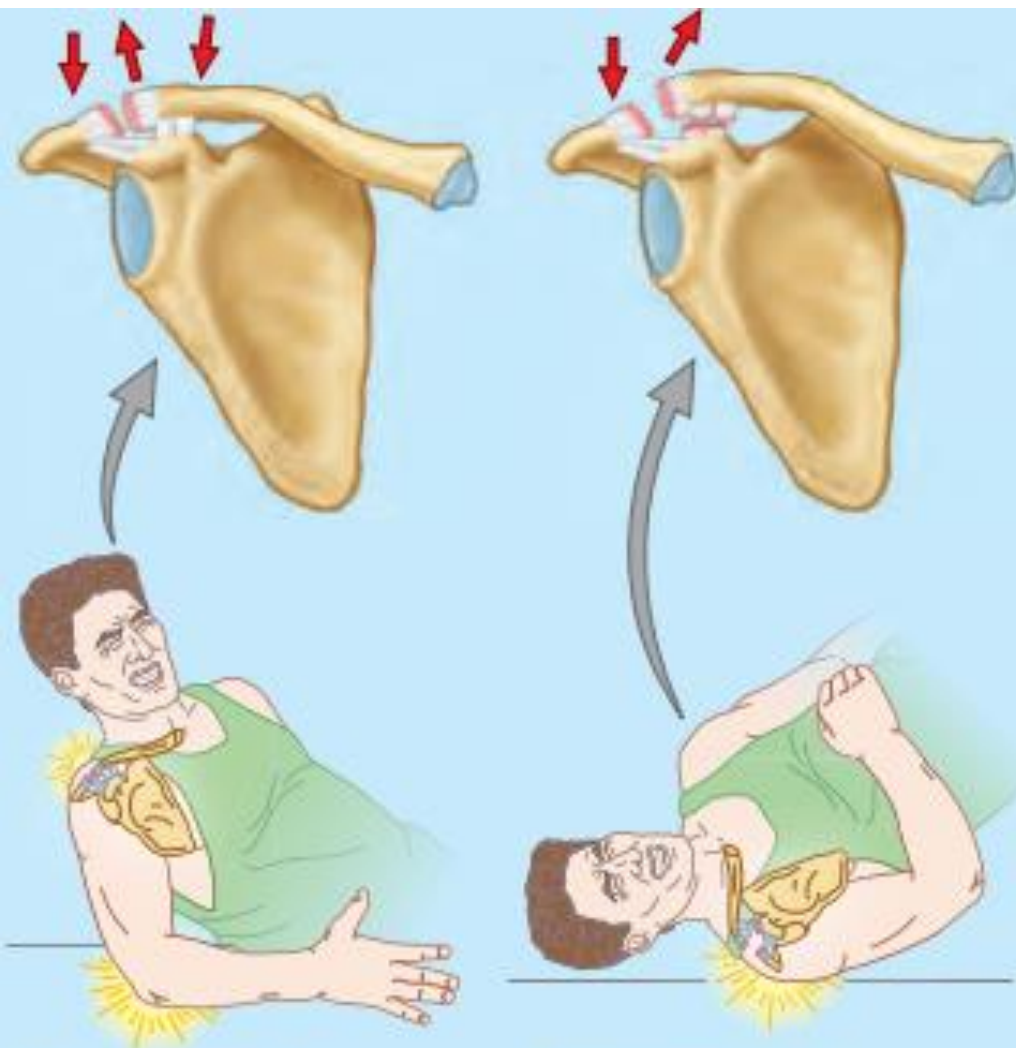
- anchors clavicle to coracoid process of the scapula
 - maintains its integrity and prevents the acromion from being driven under the clavicle when the AC joint is separated

- **Blood supply:** suprascapular and thoracoacromial arteries

- **Innervation:** Supraclavicular, lateral pectoral, and axillary nerves

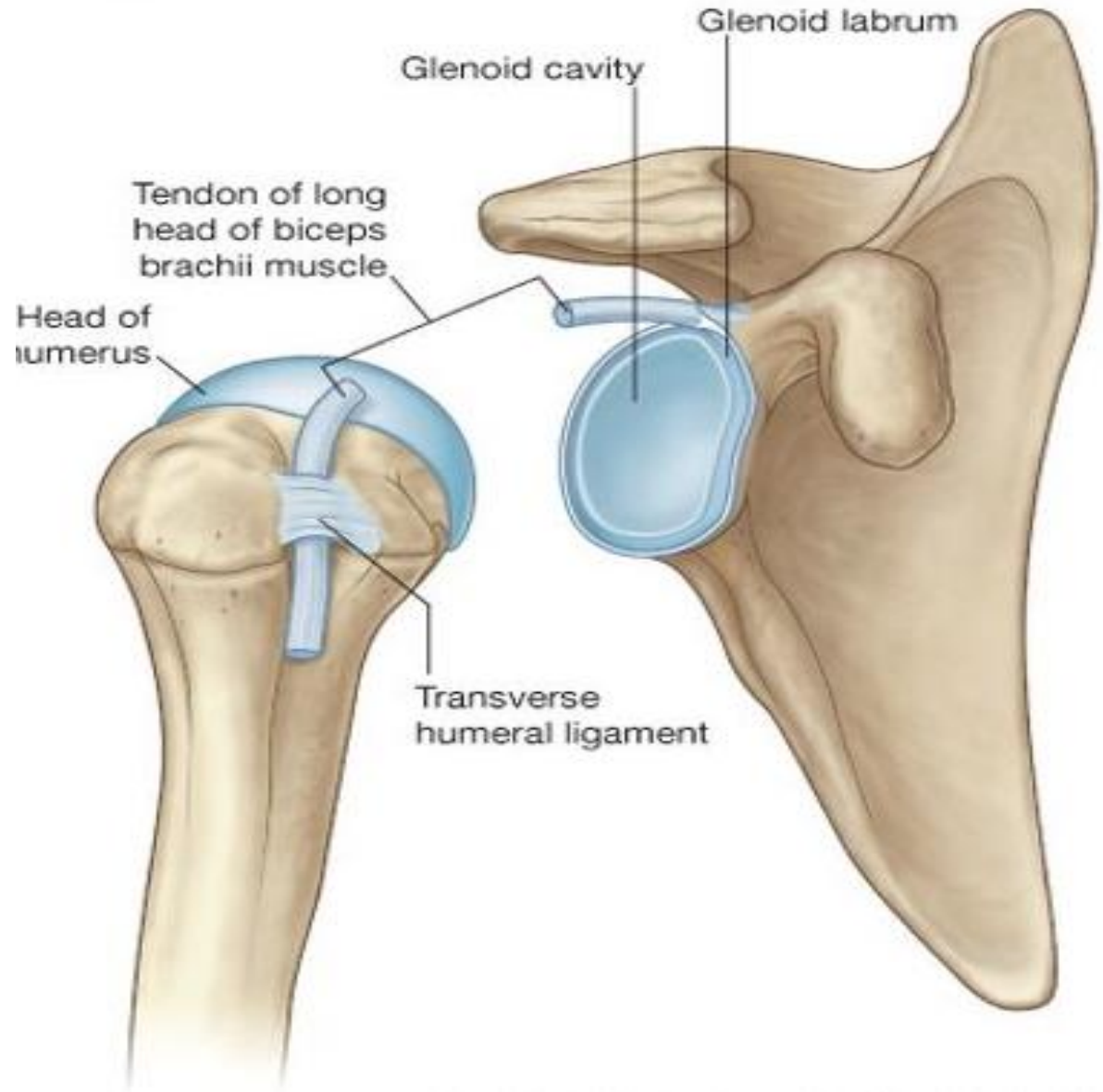
Dislocation of AC Joint

- Although its extrinsic (coracoclavicular) ligament is strong, the AC joint itself is weak and easily injured by a direct blow
- Result from a hard fall on the shoulder or on the outstretched upper limb
- When the coracoclavicular ligament tears, the shoulder separates from the clavicle and falls because of the weight of the upper limb
- Dislocation of the AC joint makes the acromion more prominent, and the clavicle may move superior to this process

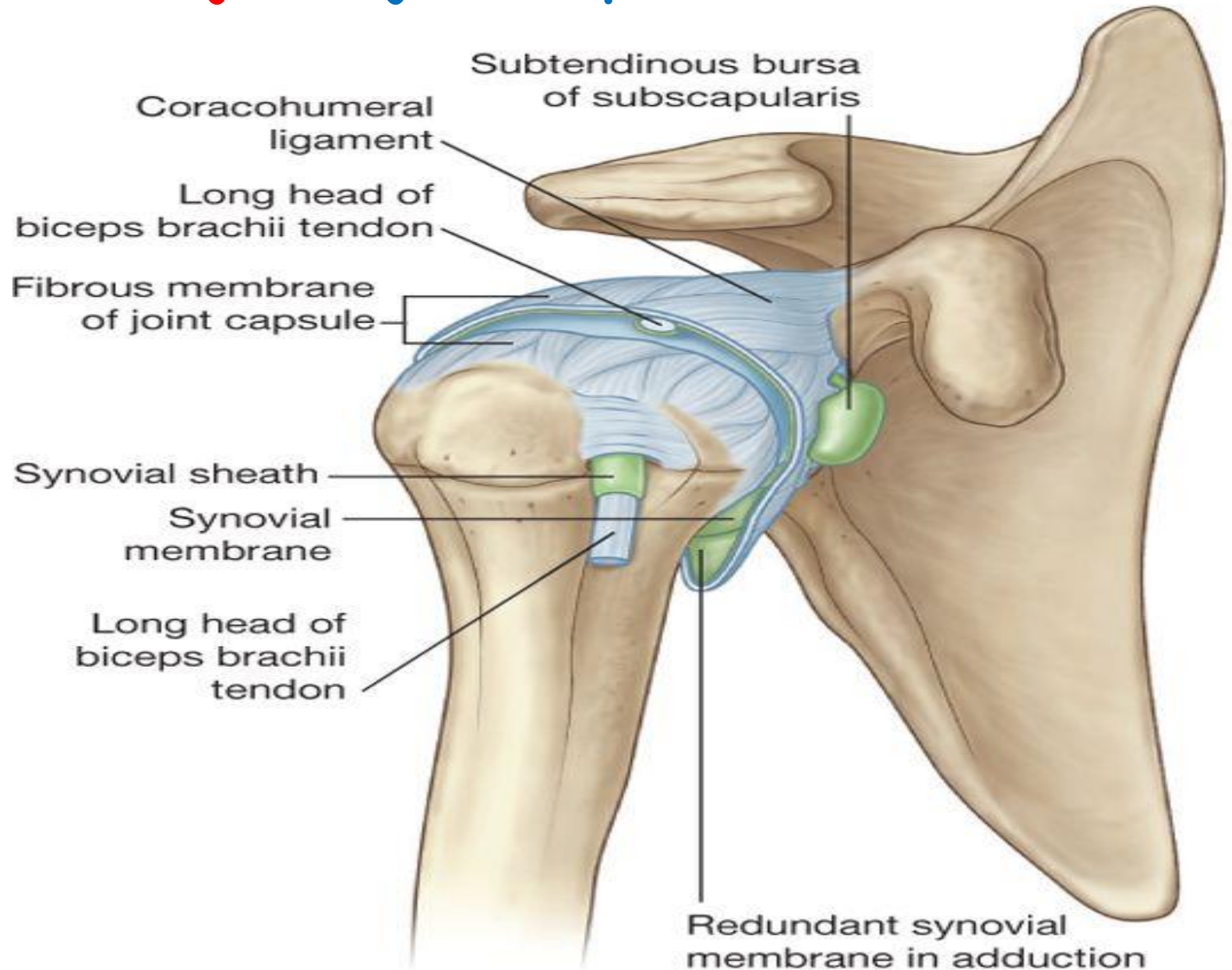


Shoulder Joint (glenohumeral joint)

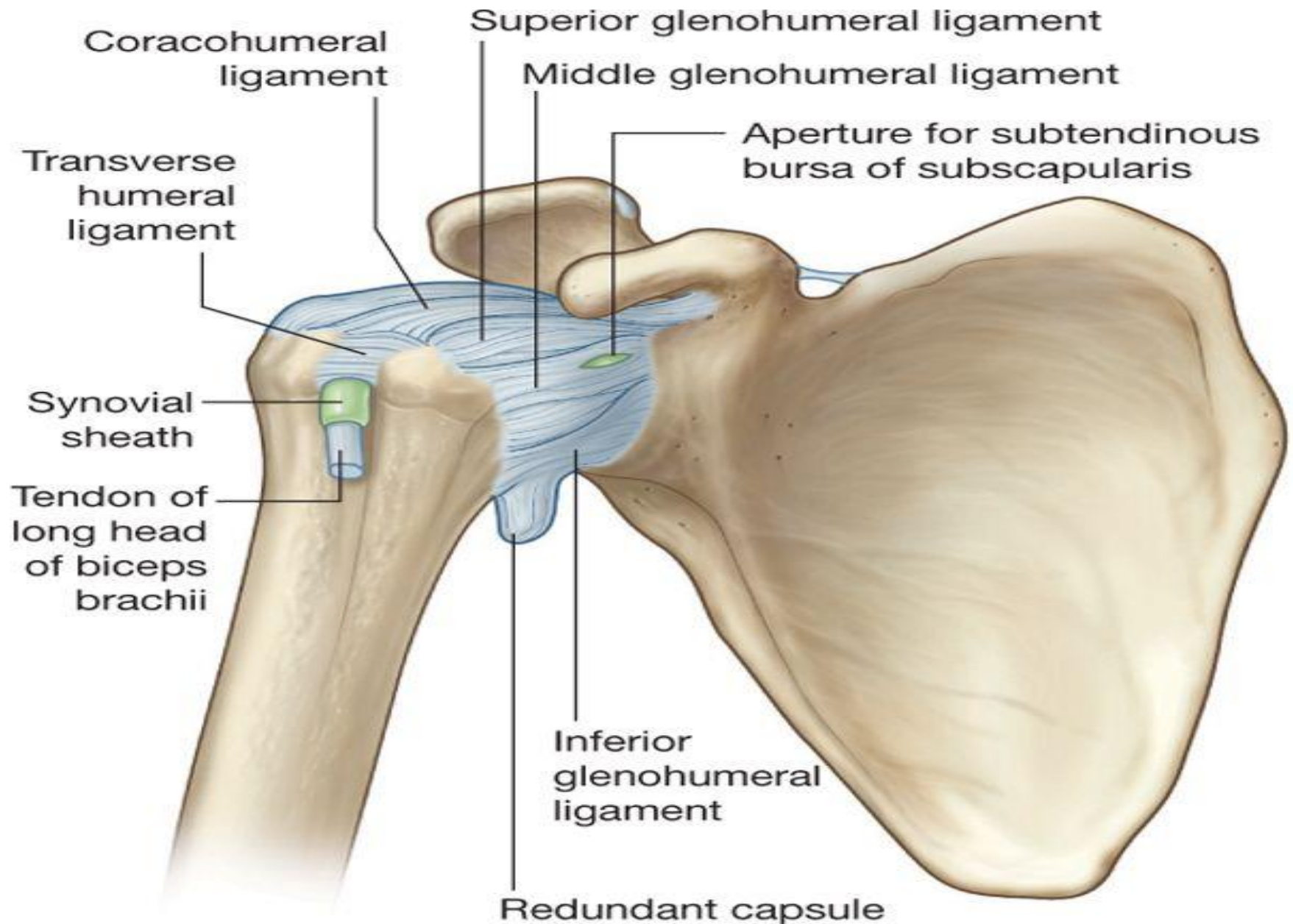
- Type
- Articulations
- Cartilages



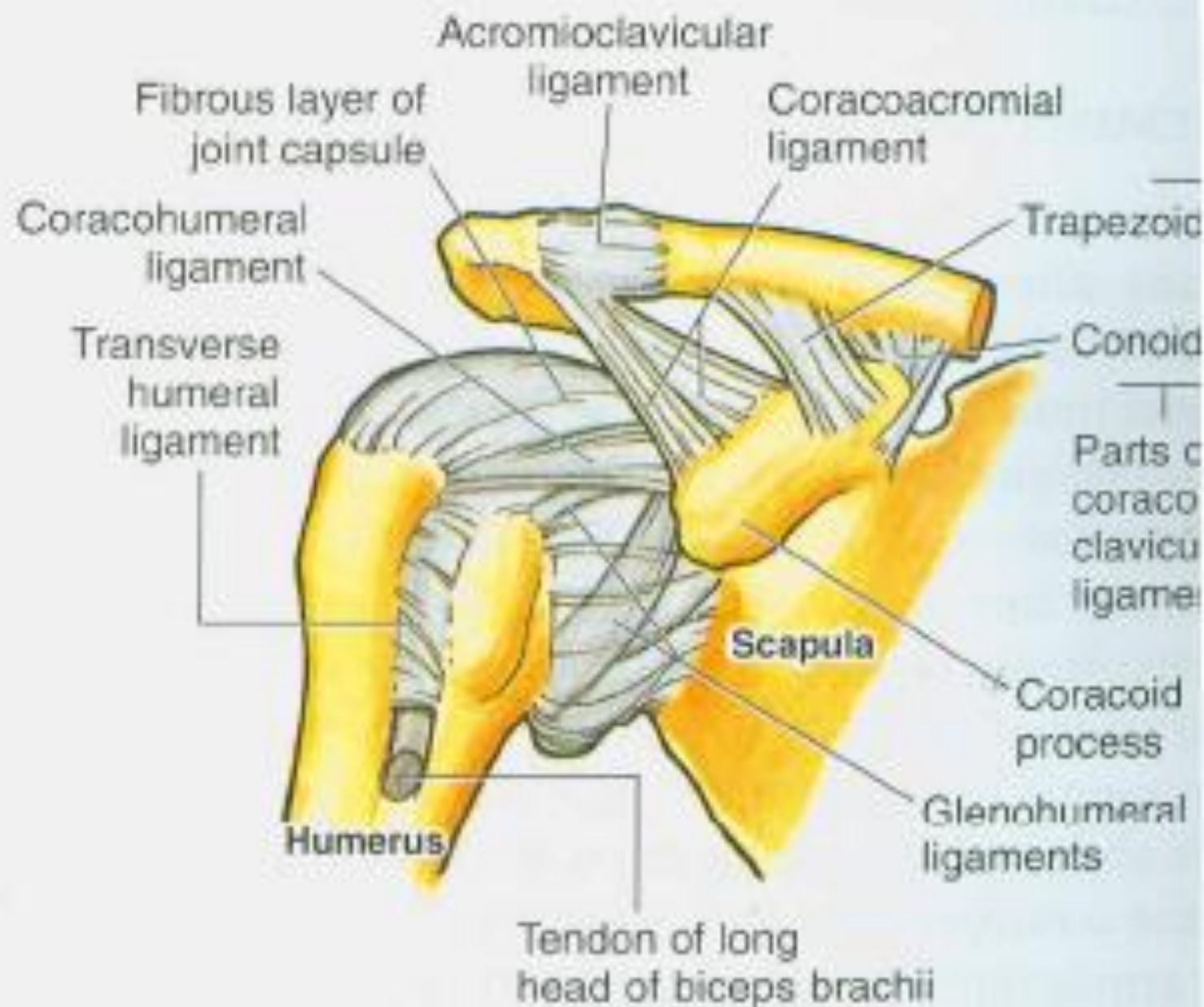
Glenohumeral joint: joint capsule



Glenohumeral joint: Ligaments

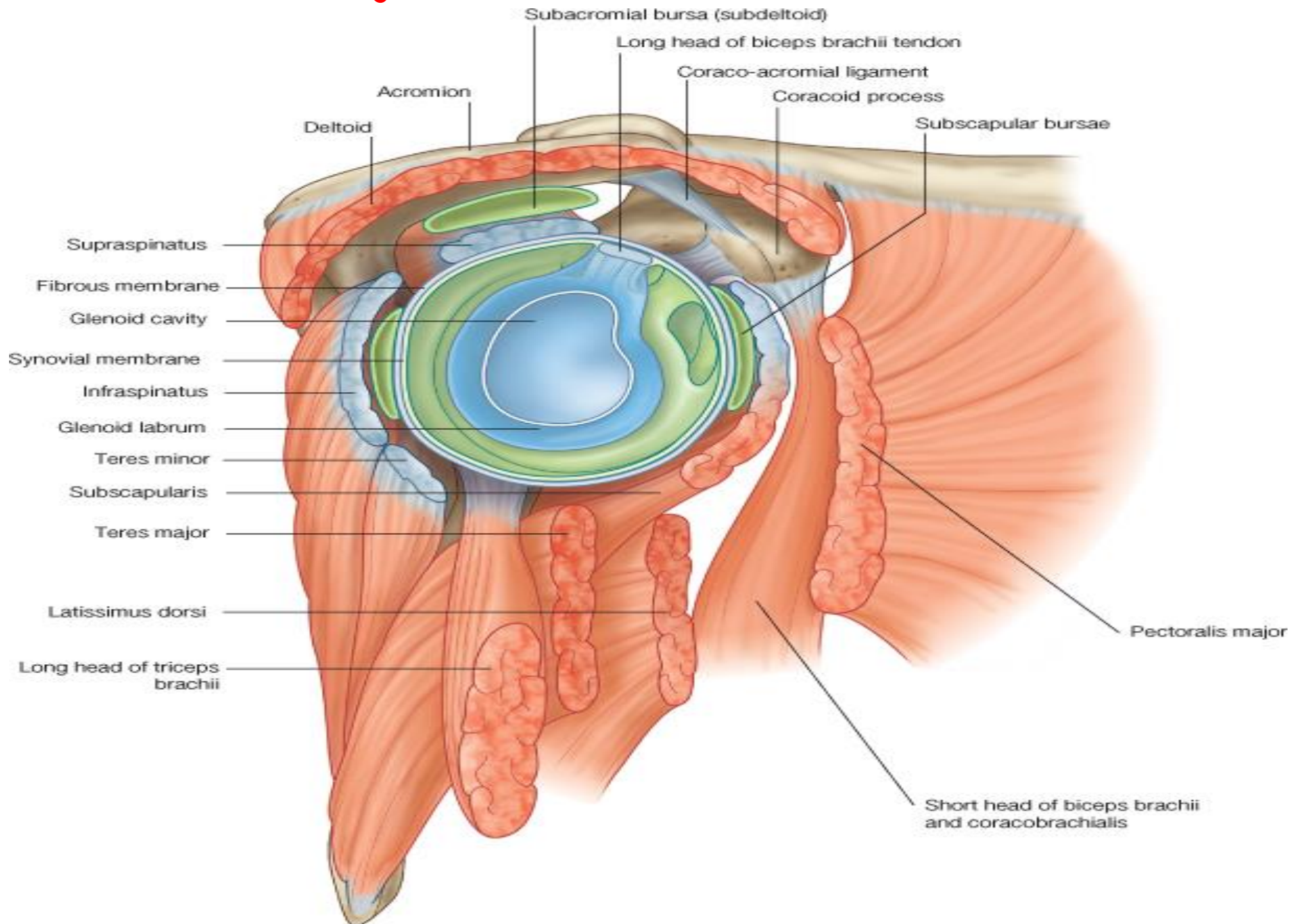


- **Ligaments**
 - **Glenohumeral ligament**
 - Strengthen anterior aspect of the capsule
 - **Coracohumeral ligament**
 - Strengthen superiorly
 - **Coracoacromial arch**
 - Formed by inferior aspect of acromion and coracoid process
 - Coracoacromial ligament spanning between them
 - Overlies the head of the humerus, prevents its superior displacement



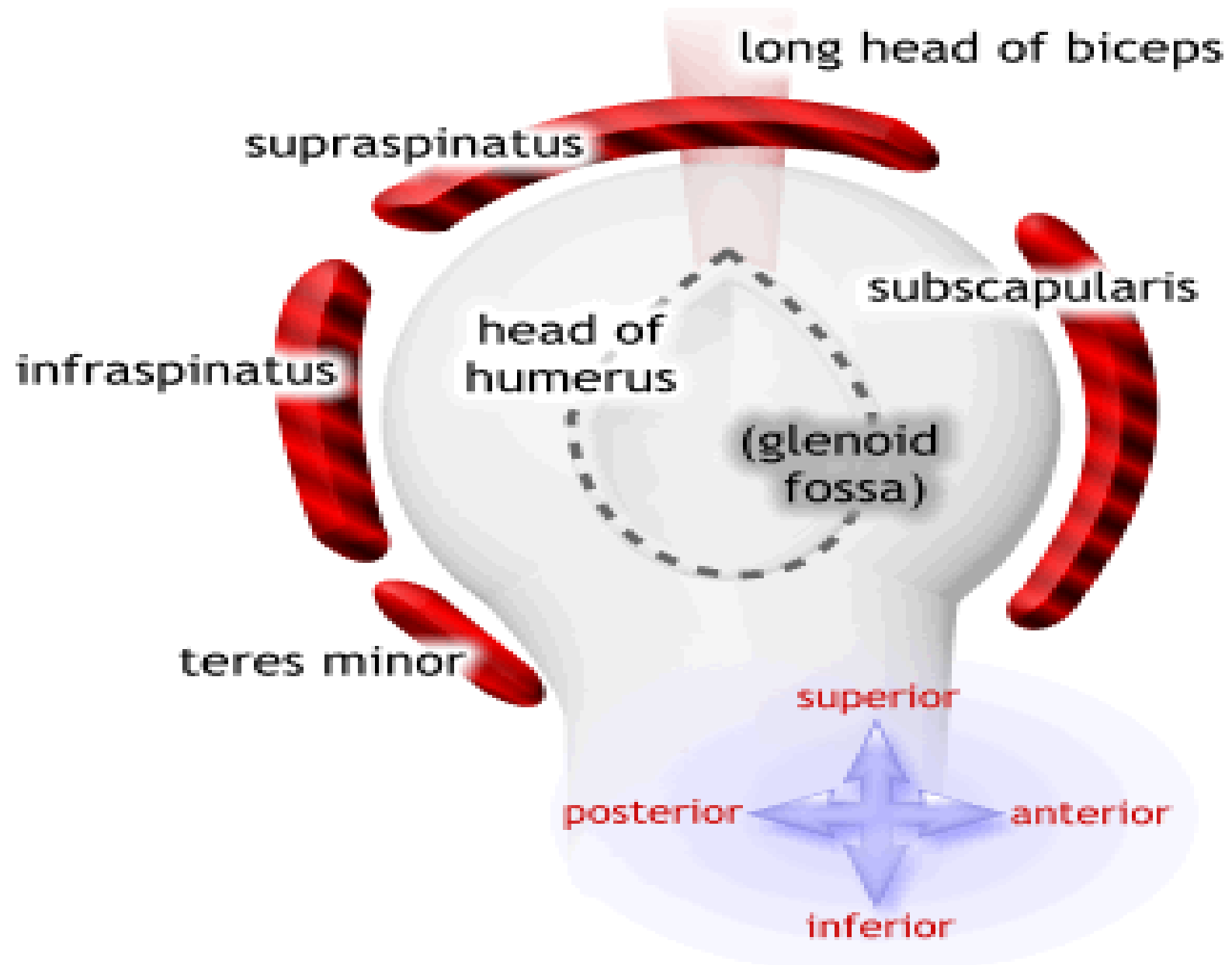
D Anterior view

Glenohumeral joint: bursae



- **Bursae around the joint**
 - **Subacromial bursa**
 - located between the acromion, coracoacromial ligament and deltoid superiorly and the supraspinatus tendon and joint capsule of the glenohumeral joint inferiorly
 - **Subscapular bursa**
 - located between the tendon of the subscapularis and the neck of the scapula

Glenohumeral joint: muscles surrounding



- **Movements**

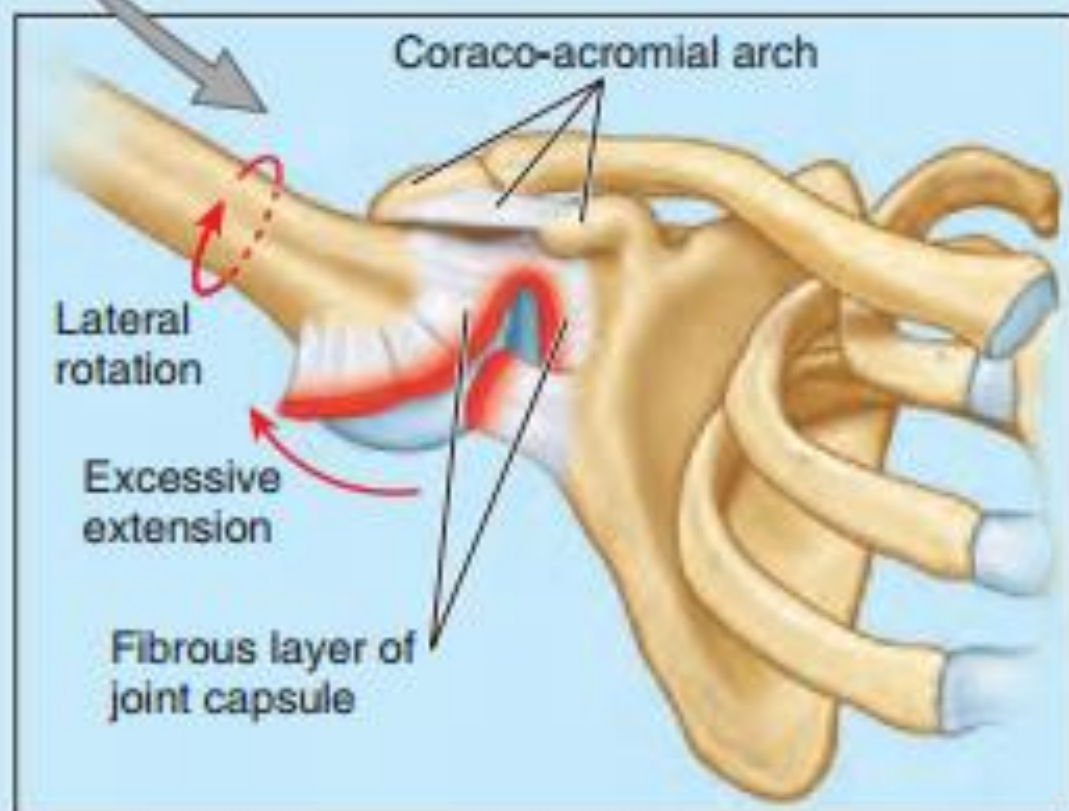
- Most movable joint in the body
- The freedom of movement results from laxity of joint capsule and the large size of humeral head compared with the small size of glenoid cavity
- Movements: flexion-extension, abduction-adduction, rotation and circumduction

- Blood supply: anterior and posterior circumflex humeral arteries and branches of suprascapular artery

- Innervation: suprascapular, axillary and lateral pectoral nerves

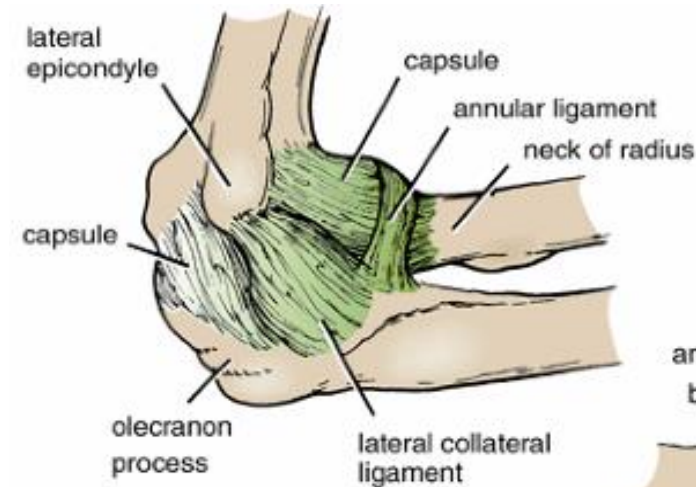
Dislocation of the Glenohumeral Joint

- Because of its freedom of movement and **instability**, the joint is commonly dislocated by direct or indirect injury
- Most dislocations of the humeral head occur in the **inferior direction**
- Usually caused by excessive extension and lateral rotation of the humerus
- A hard blow to the humerus when the glenohumeral joint is fully abducted tilts the head of the humerus inferiorly onto the **inferior weak part** of the joint capsule
- This may tear the capsule and dislocate the joint so that the humeral head comes to **lie inferior**
- Unable to use the arm, the person commonly supports it with the other hand
- The **axillary nerve** may be injured when the glenohumeral joint dislocates because of its close relation to the inferior part of the capsule of this joint

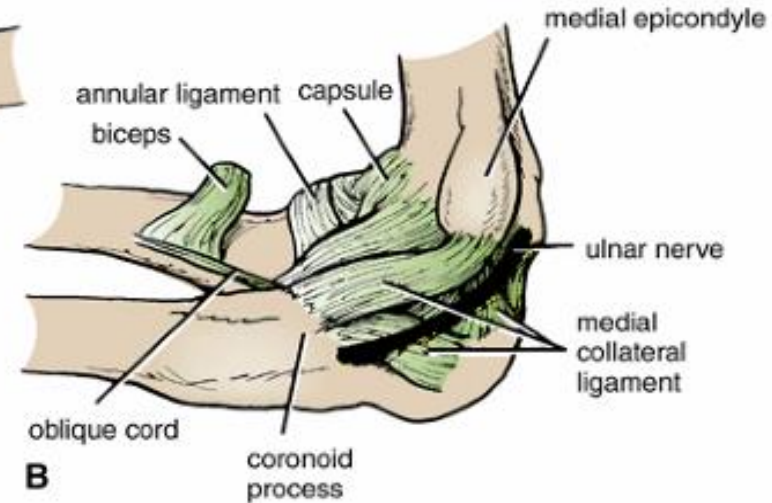


Elbow Joint

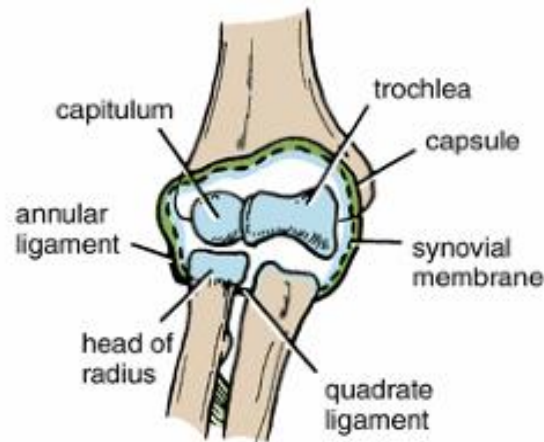
- Type
- Articulations
- Cartilages
- Articular capsule
- synovial membrane



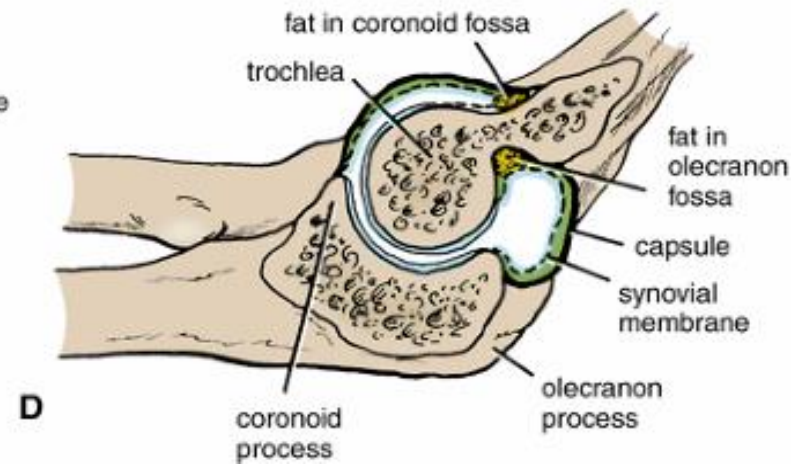
A



B



C



D

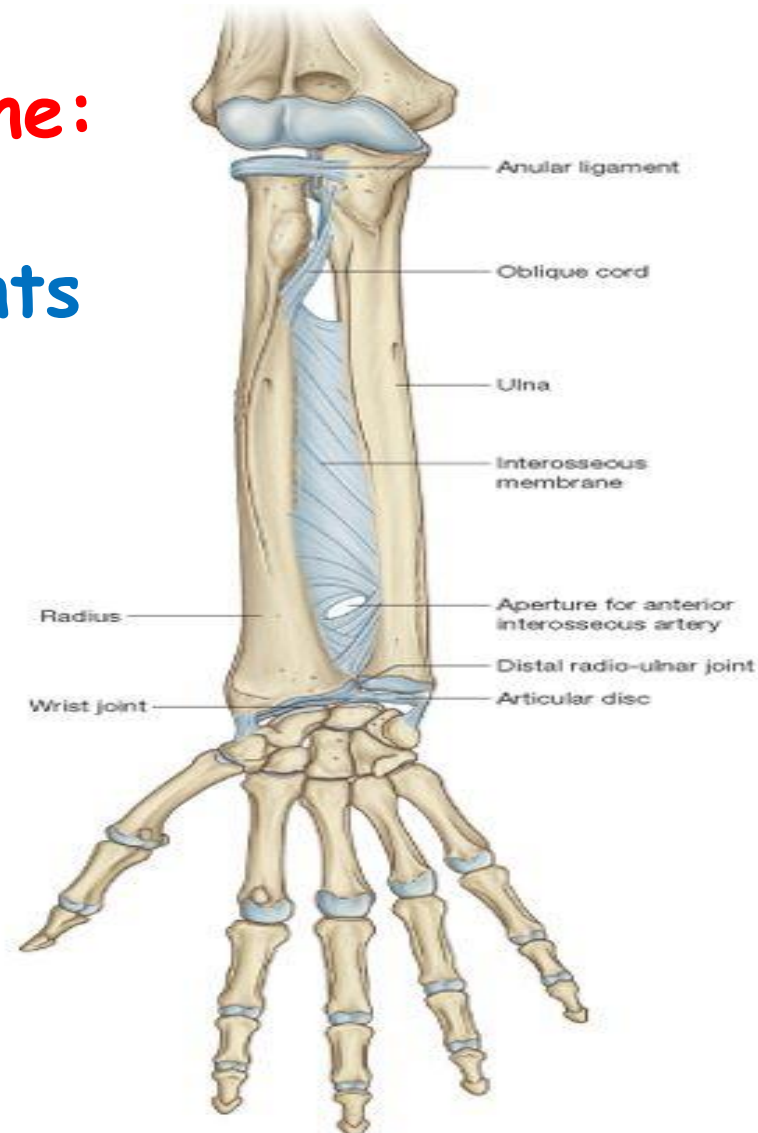
- **Ligaments**
 - **radial collateral ligament**
 - extends from the lateral epicondyle of the humerus and blends distally with annular ligament of the radius
 - encircles and holds the head of the radius in the radial notch of the ulna, forming the proximal radioulnar joint
 - **ulnar collateral ligament**
 - extends from the medial epicondyle of the humerus to the coronoid process and olecranon of the ulna
- **Blood supply:** derived from the anastomosis of arteries around the elbow joint
- **Innervation:** musculocutaneous, radial, and ulnar nerves

Elbow Joint: movements



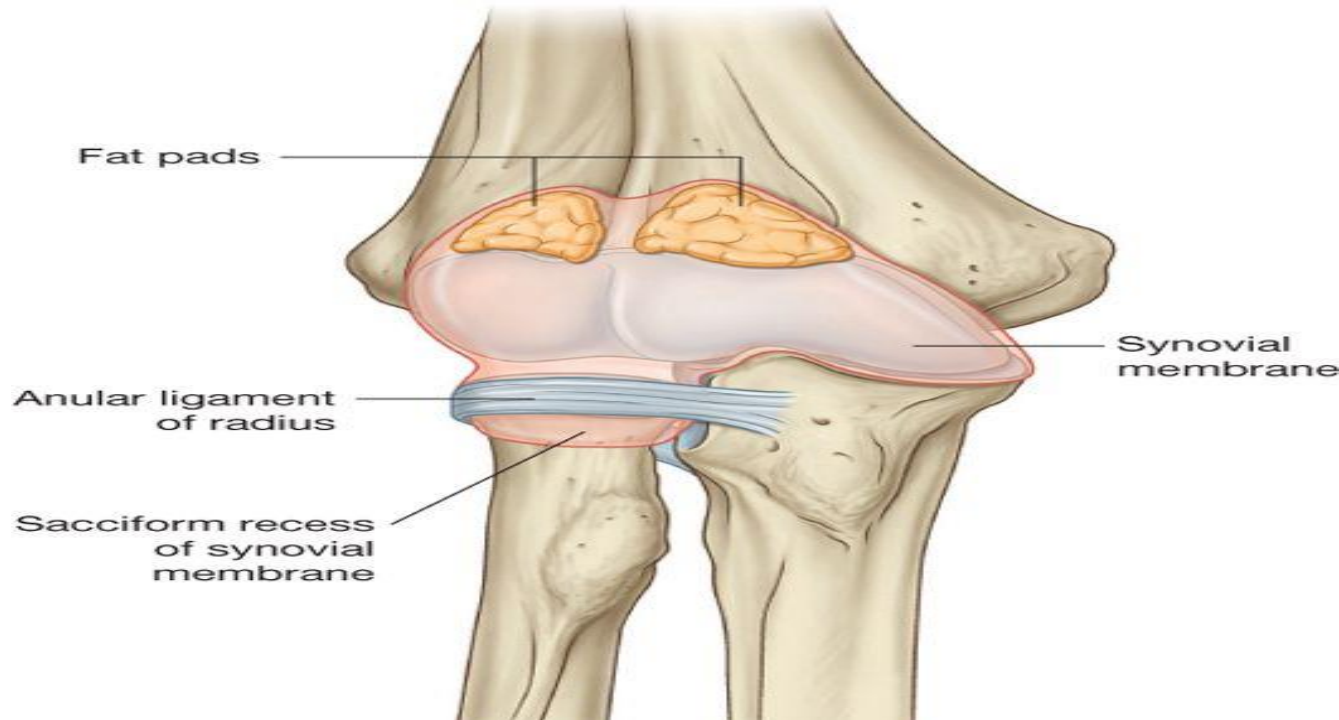
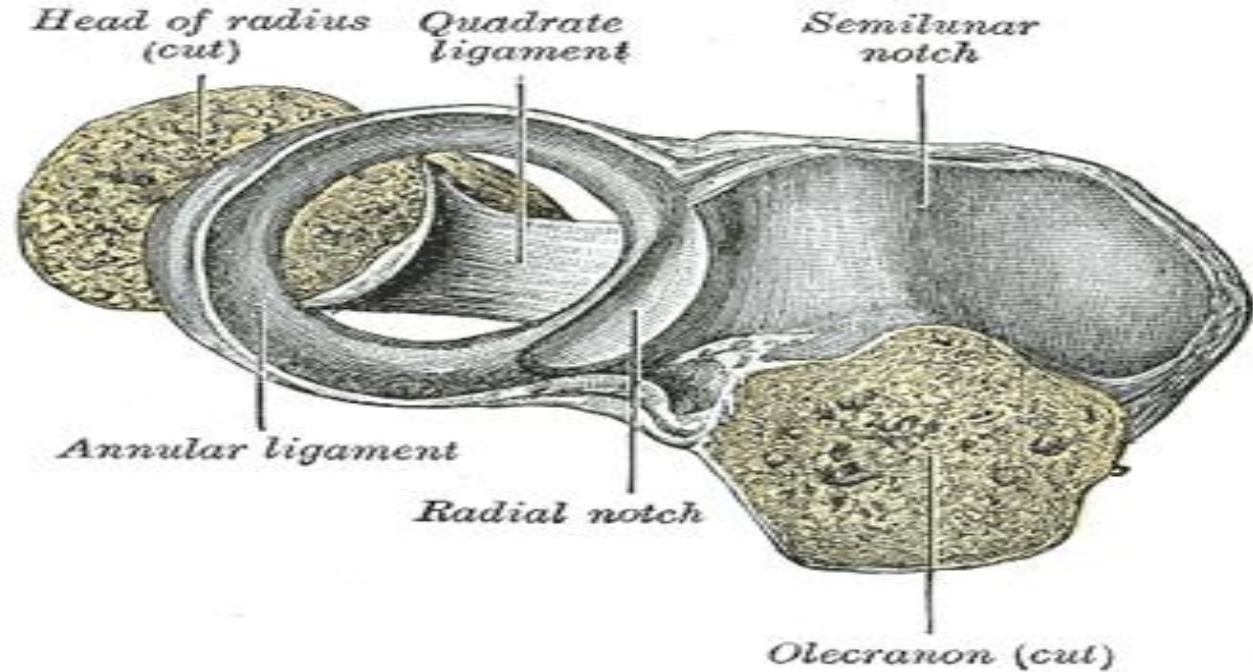
Radioulnar Joints: Proximal and distal

Interosseous Membrane:
is common to both
proximal & distal joints



Proximal Radioulnar joint

- Type
- Articulations
- Ligaments
- Articular capsule



- **Ligaments**

- **Annular ligament**

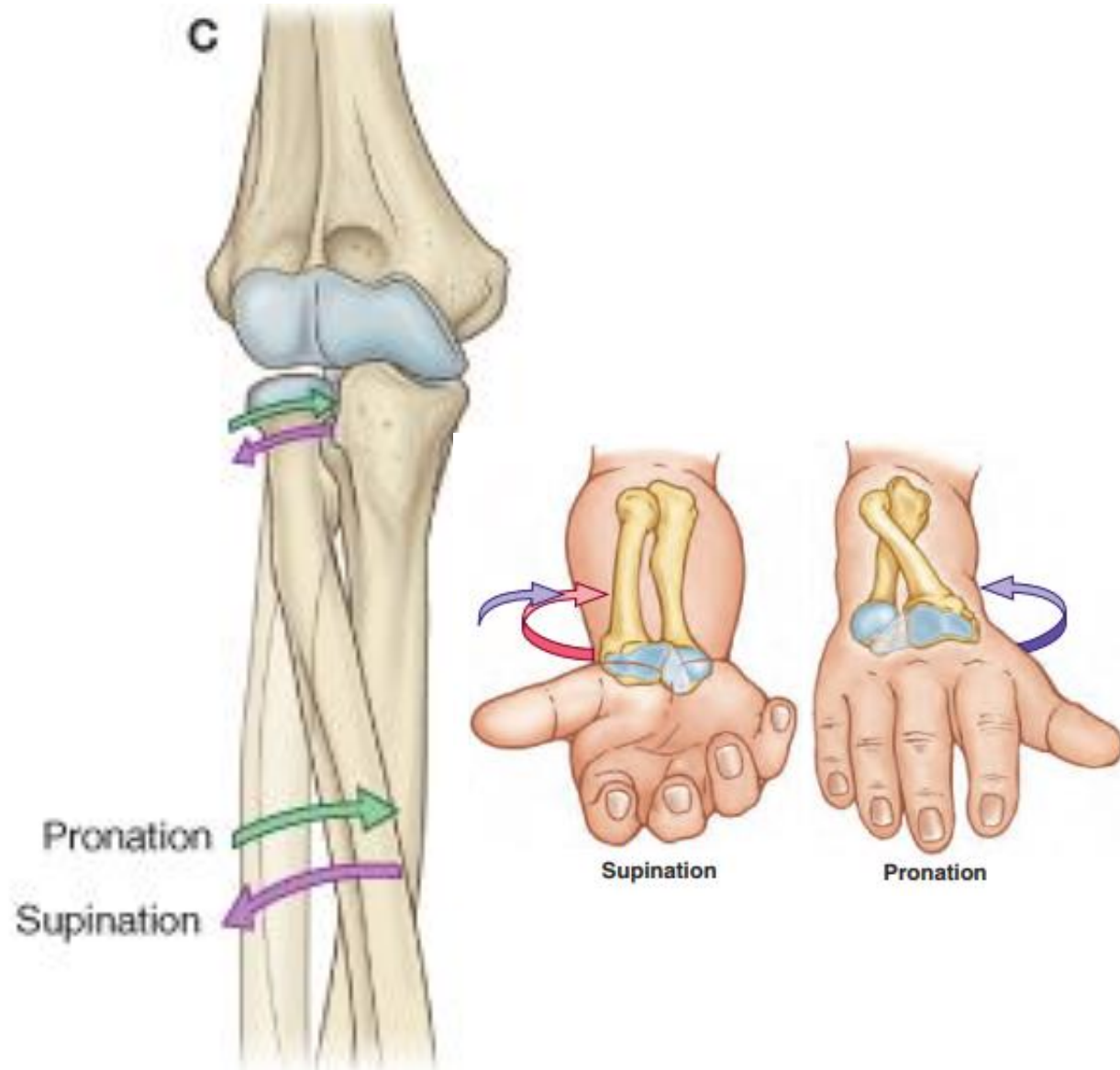
- attaches to the ulna, anterior and posterior to the radial notch, which forms a collar that, with the radial notch, forms a ring that completely encircles the head of the radius

- **Blood supply:** the radial portion of the arterial anastomosis of the elbow joint

- **Innervation:** musculocutaneous, median, and radial nerves

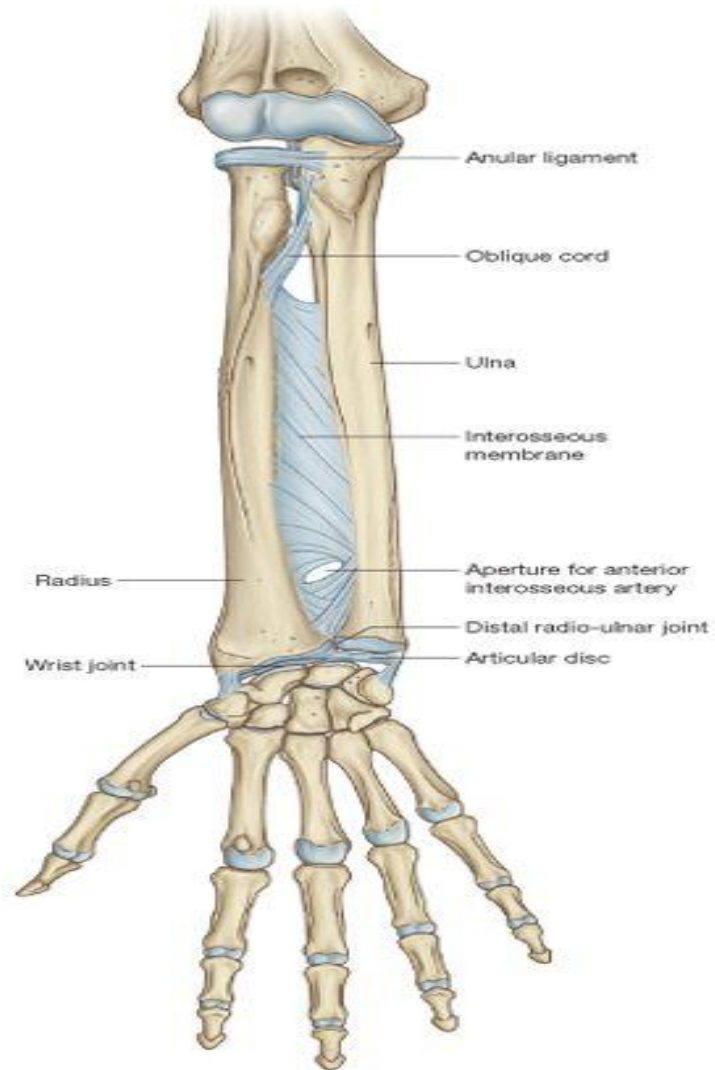
- Pronation is essentially a function of the median nerve, whereas supination is a function of the musculocutaneous and radial nerve

Proximal radioulnar joint: movement



Distal Radioulnar joint

- Type
- Articulations
- Cartilages (fibro)
- Articular capsule
- synovial membrane

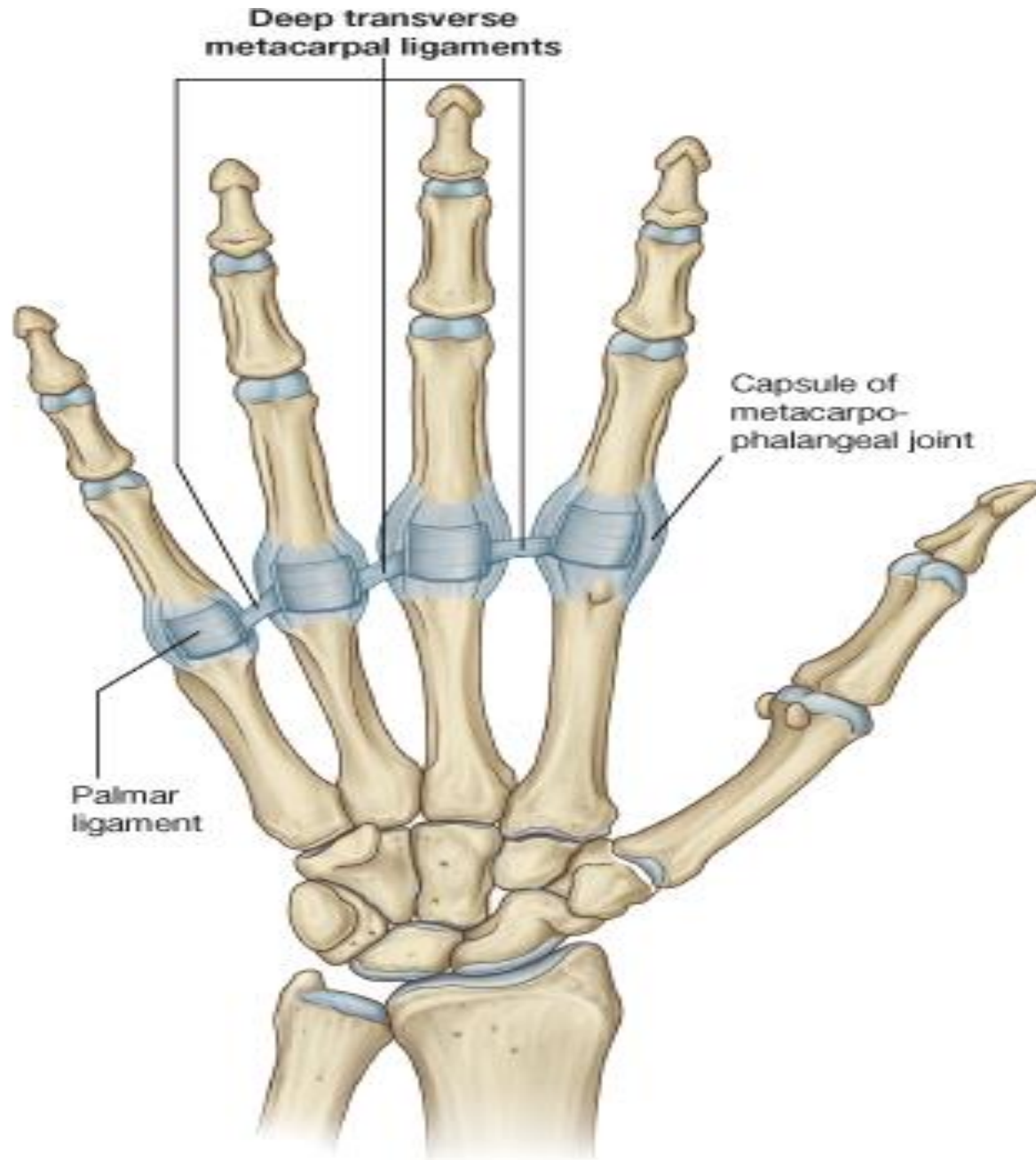


Drake: Gray's Anatomy for Students, 2nd Edition.

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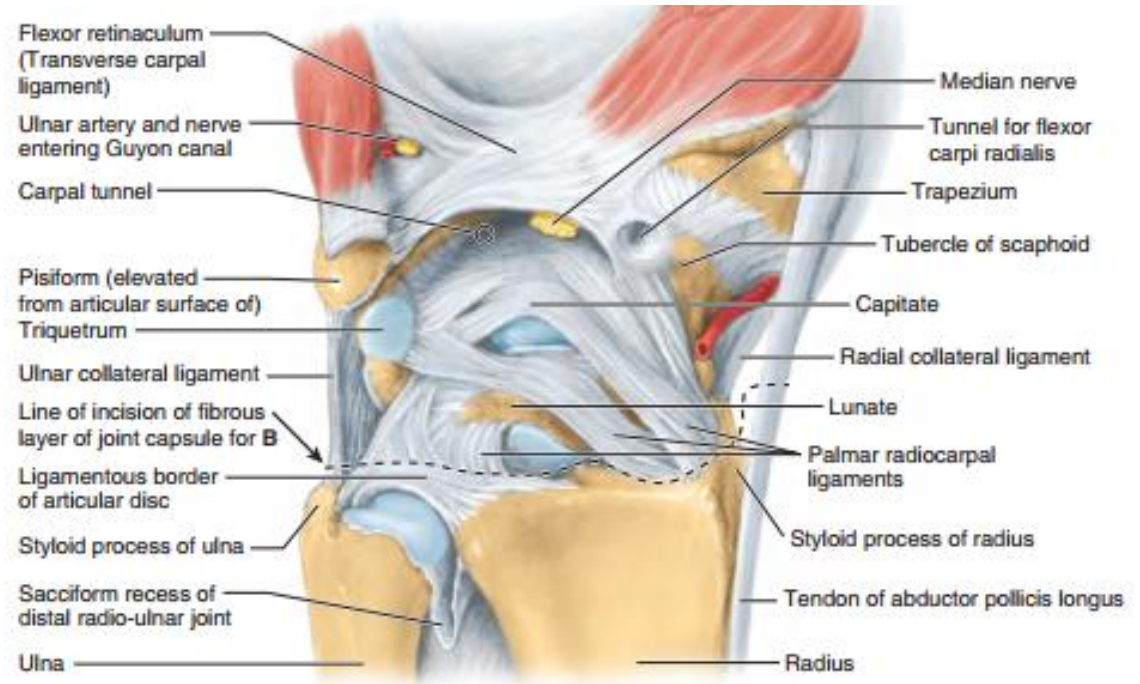
- **Ligaments**
 - Anterior and posterior ligaments strengthen the fibrous layer of the joint capsule
- **Movements**
 - The distal end of the radius rotates around the head of the ulna during pronation of the forearm and hand
- **Blood supply: anterior and posterior interosseous arteries**
- **Innervation: anterior and posterior interosseous nerves**

Joints of the Hand

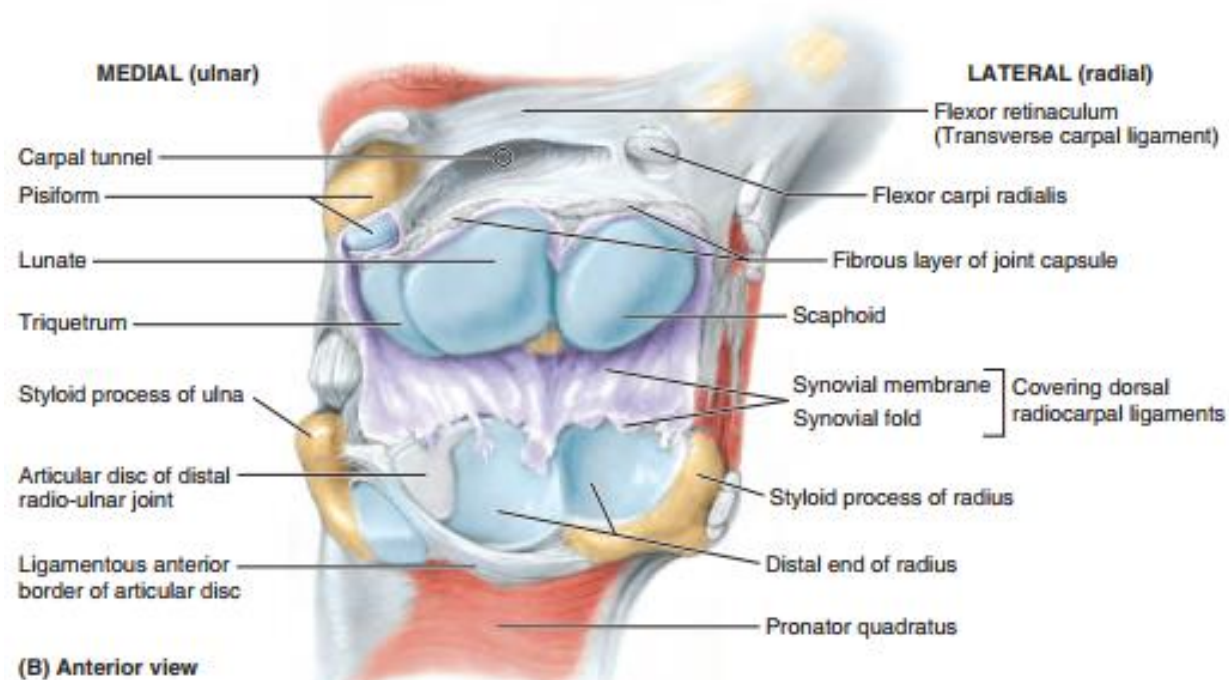


Wrist Joint (Radiocarpal Joint)

- Type
- Articulations
- Cartilages
- Articular capsule



(A) Anterior view



(B) Anterior view

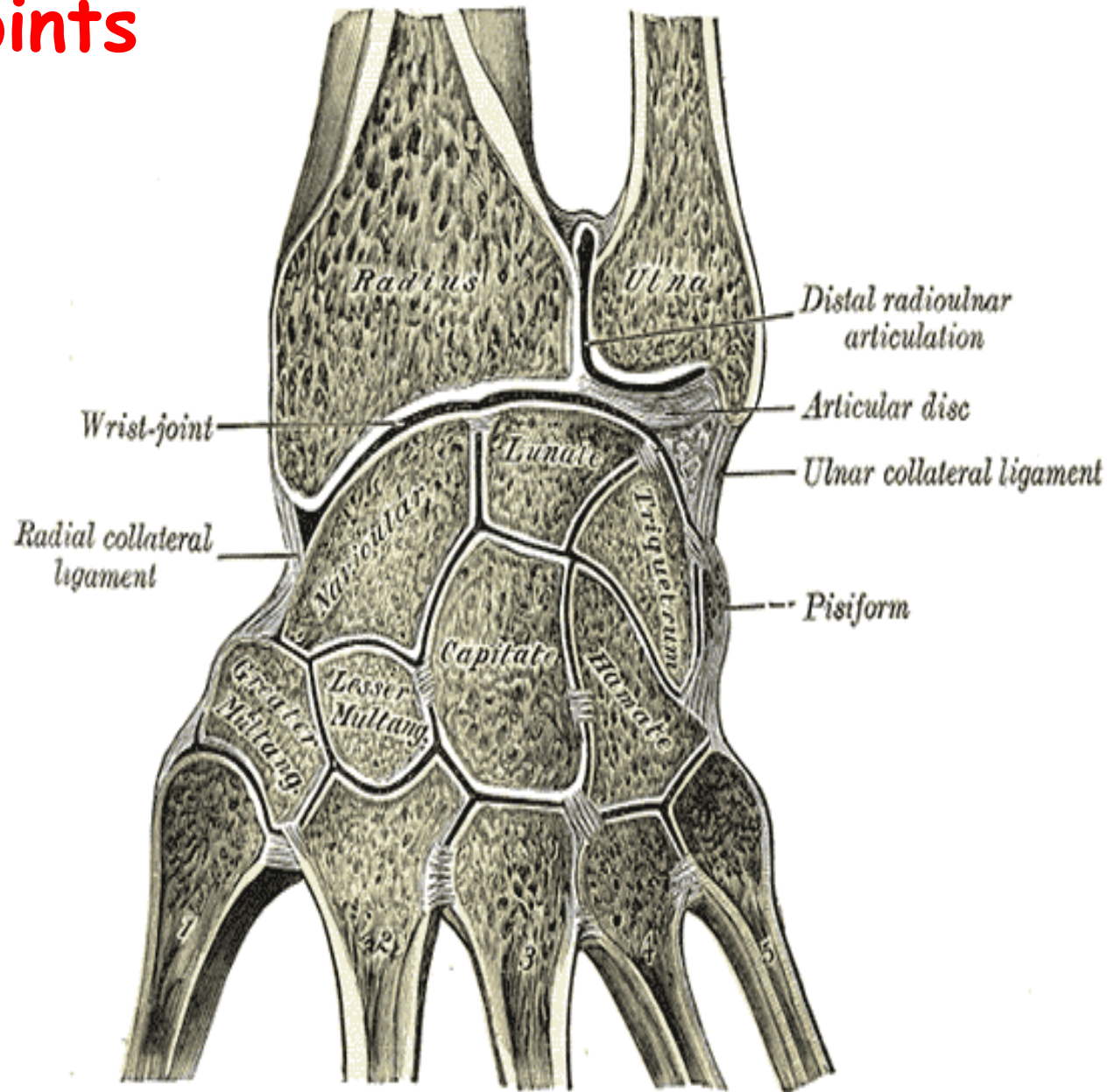
- **Ligaments**

- Anterior and posterior ligaments strengthen fibrous capsule
- Ulnar collateral ligament attaches to styloid process of ulna and triquetrum
- Radial collateral ligament attaches to styloid process of radius and scaphoid

- Movements: Flexion - extension, abduction - adduction, circumduction
- Blood supply: Dorsal and palmar carpal arches
- Innervation: Anterior interosseous branch of median nerve, posterior interosseous branch of radial nerve, and dorsal and deep branches of ulnar nerve

Intercarpal Joints (midcarpal)

- Type
- Articulations
- Cartilages
- Ligaments
- Articular capsule

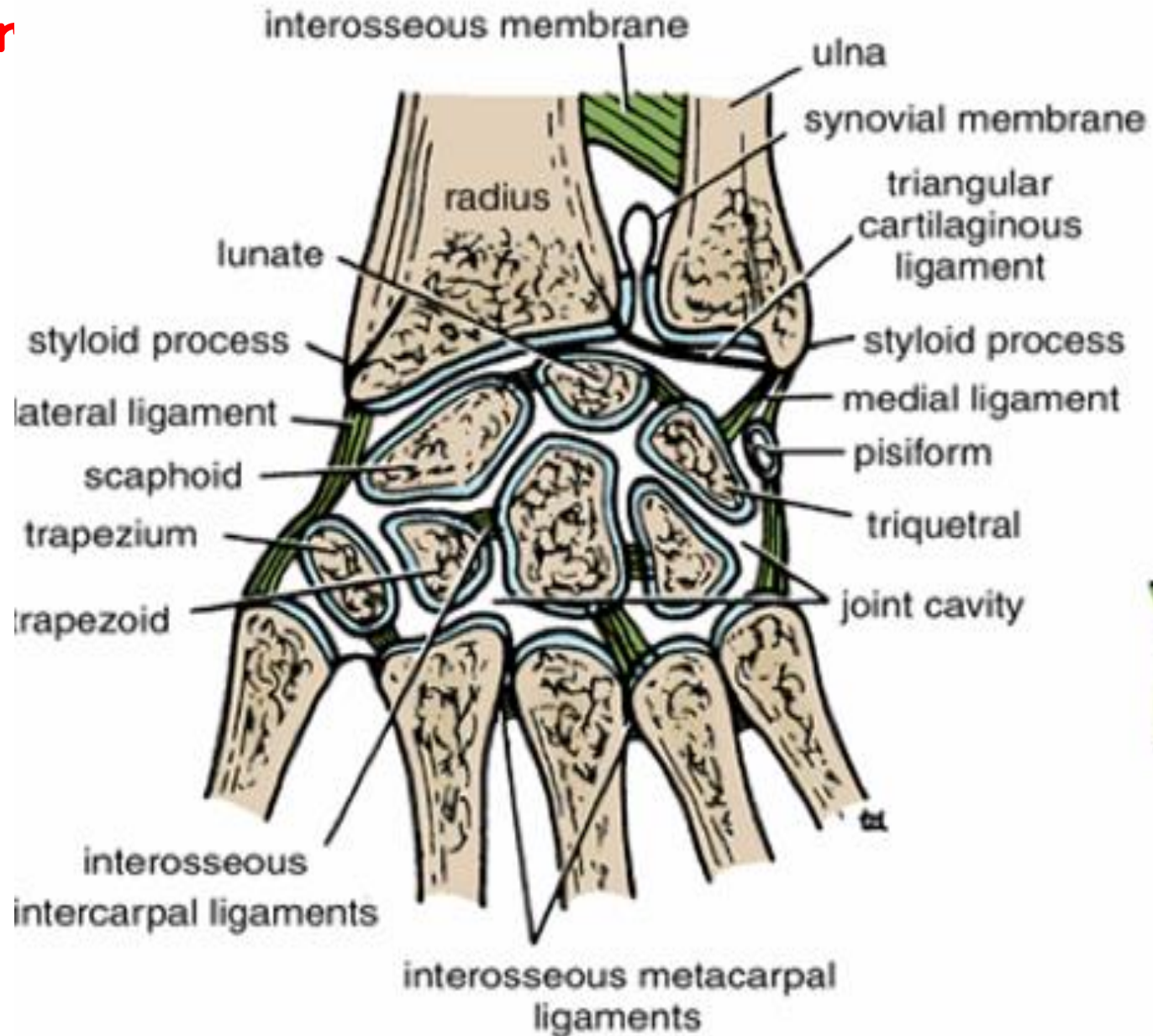


Carpometacarpal and intermetacarpal joints

- Type
- Articulations
- Ligaments

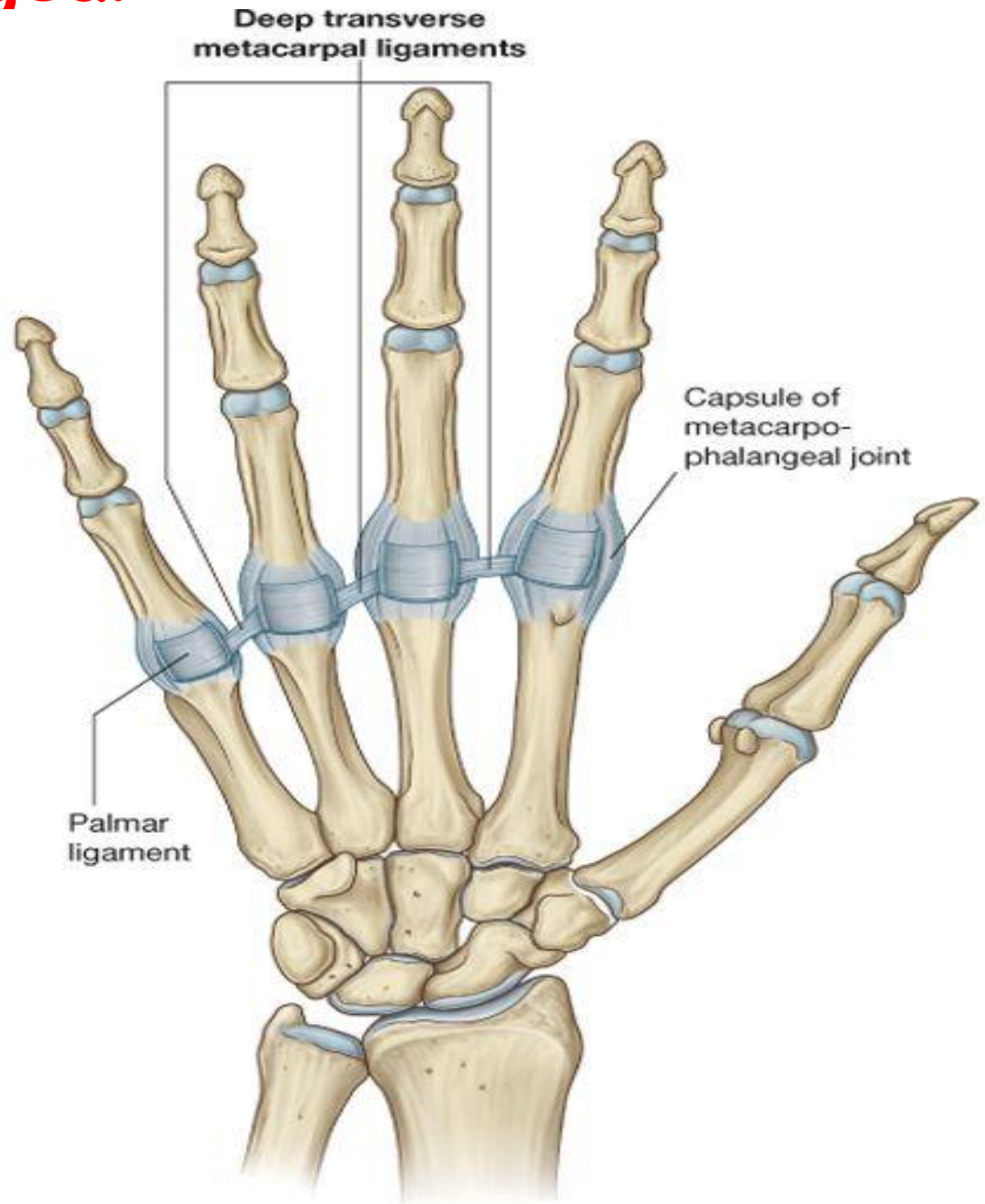
CMJ of the Thumb

- Type
- Articulation
- Articular capsule



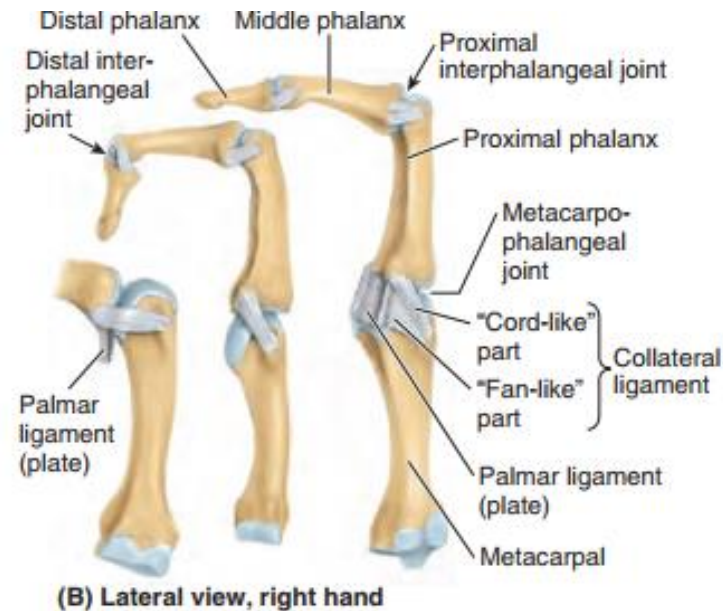
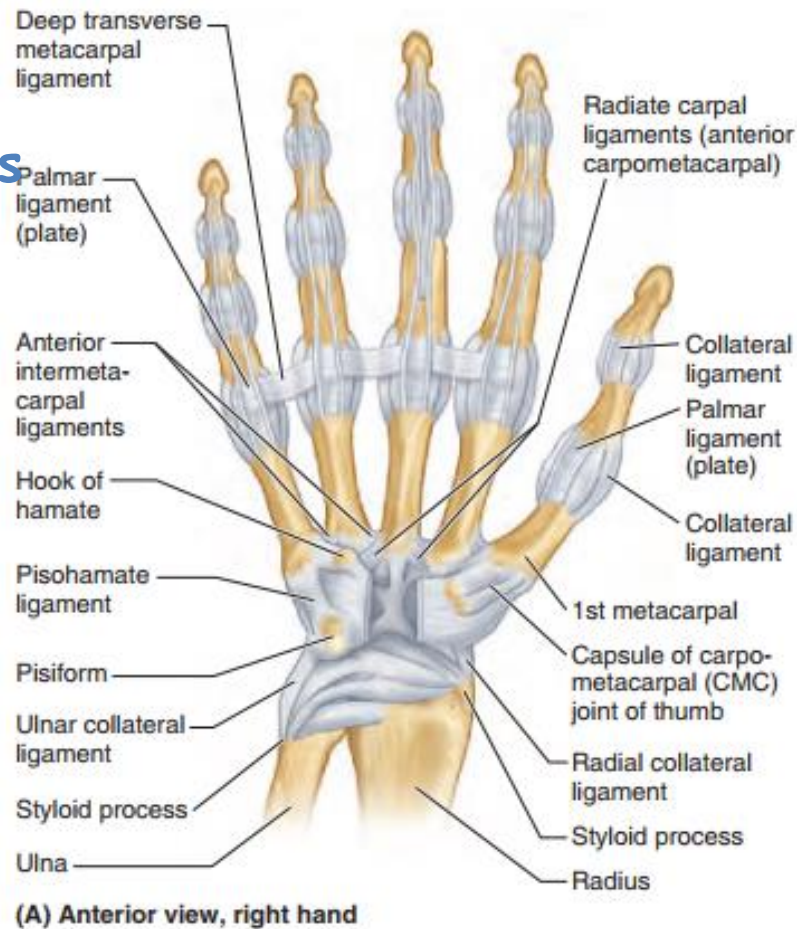
Metacarpophalangeal Joints

- Type
- Articulation
- Ligaments
- Articular capsule



Interphalangeal Joints (PIP & DIP)

- Type
- Articular capsule
- Ligaments
- Cartilages



Movements of the Hand and digits

